INCH-POUNDS

A-A-50373
1 November 1989

COMMERCIAL ITEM DESCRIPTION

VEST, COOLING (WITH FREEZEABLE GEL STRIPS)

1. SALIENT CHARACTERISTICS

1.1 Description.

1.1.1 Vest. The vest shall be made of flame retardant treated cotton, reverse twill. Six pockets (three each front and back, full width of the vest) are used to hold the freezable gel strips. The pockets are insulated with a polyolefin/polyester batting, enclosed by two layers of metallized polyolefin, and lined with nylon tricot. The pockets are fastened closed with hook and loop fastener tape. The sides of the vest are open, with four hook and loop fastener tape adjustment staps. The straps are secured by cross-box stitching. All outer edges of the front and back are bound with basic material. All seams shall be double needle stitched, except binding.

1.1.2 Freezeable gel strips. Each vest requires six freezeable gel strips which have been double bagged in polyethylene.

1.2 Components of vest.

1.2.1 Basic material. The material for the front, back, outer layer of pockets, adjustment straps and binding shall be preshrunk 100 percent cotton, reverse twill fabric. The minimum weight of the fabric shall be 10 oz. per square yard and shall be able to meet flame resistance requirements of NFPA-701. The color shall be khaki. The fabric shall not shrink more than 2 to 3 percent.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 8415

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1.2.2 Lining material. The material for lining the inner layer of the pocket shall be nylon tricot, weighing approximately 5-1/2 oz. per square yard. The color shall be white, natural or bleached white.

1.2.3 Hook and loop fastener tape. The hook and loop fastener tapes for the adjustment straps and pocket openings shall conform to the following:

Hook tape - 6.5 mil (220 denier) nylon hook with non-melting aramid ground.
Loop tape - non-melting aramid.

The color shall approximate the basic material. The hook and loop fastener tape for the pocket openings shall be 5/8 inch wide and for the adjustment straps 1-1/2 inches wide.

1.2.4 Thread. The thread for seaming and stitching the vest shall be a spun staple aramid thread, nominal 40 to 45 Tex. The color shall approximate the basic material.

1.2.5 Insulating material. The insulating material within the pocket assembly shall consist of a layer of 1/16 inch moisture resistant 65% polyolefin/ 35% polyester batting, enclosed within two layers of metallized polyolefin. The non-metalized surface shall be against the batting layer (see 4.3).

1.3 Gel strips. Nine ounces of plastic like gel (semi-solid starch gel) shall be sealed into a 0.004 mil low temperature polyethylene bag. The plastic like gel shall be able to withstand temperatures below -20 degrees centigrade and shall not leak from the bags if punctured. The seams shall be heat sealed and the pack shall measure approximately 5 by 5 inches. Three gel bags shall be sealed into a strip with heat sealed seams along the four sides and between each bag. When a tub bag is used, the top and bottom does not have to be sealed. The gel strips shall measure approximately 6 by 16 inches. The polyethylene material for both the inner and outer pack shall be a minimum of 0.004 mil ($\pm$ 20% tolerance) thickness.

1.4 Size. The vest shall be in one size only. Adjustment shall be made by hook fastener tape attached to the side adjustment straps matching with the loop fastener tape stitched to the top of the two lower front pockets.

1.5 Patterns and pattern parts. The Government will furnish a pattern for the one size. The following is a list of the pattern parts and number of cut parts:

Nomenclature	Cut parts
Front	1
Front pkt 1/	1
Back	1
Back pkt 1/	1
Strap	8

1/ The pattern for the nylon lining and insulating material shall be the same at the front and back pockets.

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1.6 Labels. A commercial type label is acceptable provided that additional printed cloth labels containing the information in 1.6.1 and 1.6.2 are included.

1.6.1 Contract No. and National Stock No. The vest shall have a label containing the contract number and national stock number caught in the stitching of the commercial label or stitched in the neck edge binding at the inside back.

1.6.2 Instruction label. The vest shall have an instruction label for donning and cleaning, containing the following information. The label shall be caught in the neck binding at the inside back.

Donning Instructions

1. Insert frozen packs into each of the pockets. Close pockets.
2. Slip vest over head. Adjust side straps for a close, comfortable fit. One layer of clothing shall be worn under vest.

Cleaning Instructions

Close hook and loop fastener tapes.
Machine wash warm or cold water only.
Tumble dry cool or hang dry.

DO NOT USE HEAT IN WASHING AND DRYING OF THIS GARMENT

or

Dry clean, using no heat.

1.7 Workmanship. After assembly, the vest shall be thoroughly cleaned and all loose threads, lint and foreign matter removed. The vest shall be uniform in quality and shall be free from irregularities or defects which adversely affect performance, reliability or durability.

1.8 Approved styles. The following styles meet the requirements of this commerical item description:

Steele Inc.
Vest: Model No. SA 1140FR
Pack: Model NO. SA 1149

Other manufacturers may submit their styles or models to the procuring activity for evaluation by the specification preparing activity, and ultimate inclusion in this document, if approved.

1.9 Product demonstration models. During the solicitation phase, interested firms (including the manufacturers of the above approved models), shall provide the procuring agency with two product demonstration models of the item they propose to furnish for evaluation. The two samples provided by the successful offerer shall be tagged and sealed. One shall be retained by the procuring agency and the other furnished the assigned QAR for use as a reference in performing the examination in paragraph 2.2 and the scoring of defects in design, material, and workmanship not otherwise classified herein.

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2. QUALITY ASSURANCE

2.1 Certification. When requested by the contracting officer, the contactor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

2.2 Examination of the end item. The vest shall be examined for the defects listed below. The lot size shall be expressed in units of one completed vest. The sample unit shall be one vest. The inspection level shall be II and the AQL's shall be 1.5 major and 10.0 for major and minor combined.

Defects		Classification Major	Minor
I	MATERIAL DEFECTS AND WORKMANSHIP DAMAGE		
	A. Component part full, tight or twisted	X	
	B. Open or puckered seam	X	
	C. Any size hole, cut, tear, smash, or misweave	X	
	D. Missing yarn or broken yarn		X
	E. Loose yarn		X
	F. Visible mend, needle chew or irregular stitching		X
	G. Pleat or crease		X
II	SHADE VARIATIONS		
	A. Parts badly shaded from other parts or within parts		X
	B. Thread color not a good match with basic material		X
III	CLEANNESS		
	A. Any spot or stain clearly noticeable		X
	B. Thread ends not trimmed		X
IV	DESIGN AND ASSEMBLY		
	A. Not cut or fabricated in accordance with the approved style/product demonstration model	X	
V	SEAMS AND STITCHING		
	A. Seams puckered, twisted, pleated or any part of vest caught in an unrelated operation	X	
	B. Any open seam, loose stitch tension or seam, stitch type and stitches per inch not in accordance with approved style/product demonstration model		X

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Defects	Classification Major	Minor
VI GENERAL		
A. Any defect not otherwise classified herein but at variance with approved style/product demonstration model		X
VII POCKETS		
A. Not large enough for packs	X	
B. Opening does not stay closed	X	
C. Not uniform in size and location		X
VIII BINDING		
A. Binding omitted, twisted or not securely attached	X	
IX SIDE STRAPS		
A. Incorrect width or length, hook fastener tape not secured	X	
B. Strap not uniformly constructed	X	
C. Not attached with cross box stitch	X	
X LABEL		
A. Label missing, incorrect or illegible		X
XI MEASUREMENTS		
A. Any part of garment not in accordance with tolerances on figure or garment measurement not in accordance with figure.	X	

3. PACKAGING, PACKING AND MARKING

3.1 Packaging. Each vest shall have the fastener tape engaged, pockets closed and inside of front against inside of back. Fold each vest along top of middle pocket on front and back. Vest should measure approximately 15-1/2 by 21 inches. The folded vest shall be inserted into a close fitting, flat style, clear polyethylene film bag.

3.2 Packing.

3.2.1 Vest. Six vests packaged as specified in 3.1 shall be packed in a fiberboard shipping container, assembled, closed and reinforced according to type CF, class domestic, variety DW, grade 200, size 2A of MIL-B-17757. The vest shall be reversed from top to bottom within the box. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Any voids in the box shall be filled with tissue paper or commercial filler.

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3.2.2 Gel strips. Eighteen (18) of the gel strips shall be packed in a fiberboard shipping container, assembled, closed, and reinforced according to type CF, class domestic, variety DW, grade 200, size 1A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. The packs shall be packed as follow: one in length, two in width and nine in depth. Any voids within the shipping container shall be filled with tissue paper or commercial filler. The container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

3.3 Marking. The unit package, intermediate and outer container and the palletized load, as applicable, shall be marked in accordance with MIL-STD-129.

3.4 Palletization. When specified, see 4.2, vests and packs, packed as specified in 3.2, shall be palletized on a 4-way entry pallet in accordance with load type 1a of MIL-STD-147. Each prepared load shall be bonded in accordance with means C and D and F or G. Pallet patterns shall be in accordance with apendix of MIL-STD-147. The pallet shall be 4-way, type IV, class 1, size 2; or fabricated from wood group I, II, III, IV, Grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from wood groups specified in MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

4. NOTES

4.1 Intended use. The vests are intended to be worn in heat stress conditions to extend Physiological Heat Exposure Limits (PHEL's) of Navy personnel.

4.2 Acquisition requirements. The acquisition document shall specify the following:

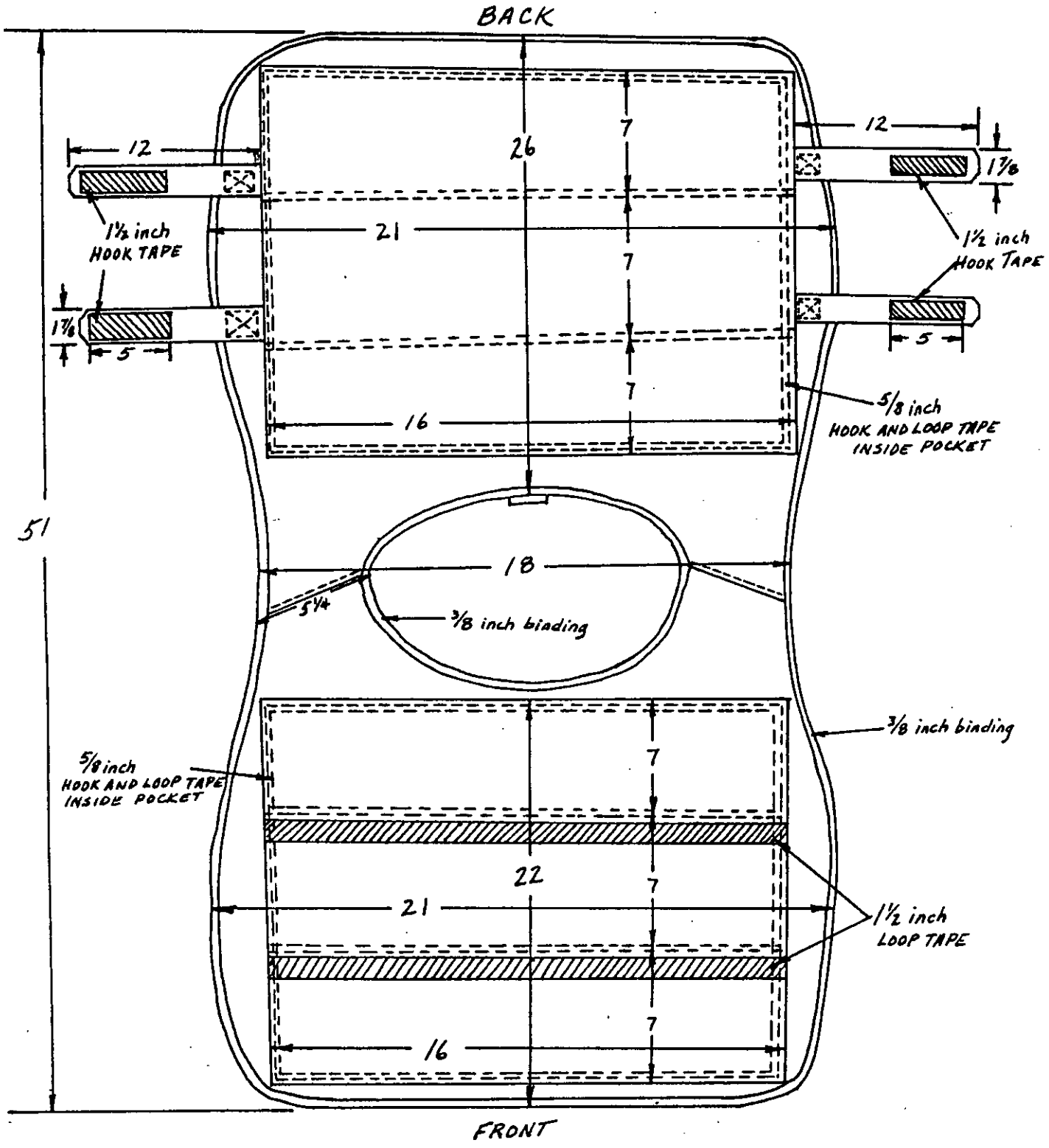
(a) Title and date of this document.
(b) When palletization is required.

4.3 Insulating material The insulating material, 65% polyolefin/35% polyester is available as Thinsulate CS 150 and the metallized polyolefin as Thermal RI from 3-M Co. of St. Paul MN 55144-1000.

Preparing Activity:
Navy - NU

Project Number:
8415-N747

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All measurements are in inches
Tolerance for all measurements is ± 5%
Hook tape should be on underside of straps as shown.

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-50373	VEST, COOLING (WITH FREEZEABLE GEL STRIPS)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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